Homework 1 Solutions IEORE4004, Fall 2025
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Problem 1

Let: x_1 be the number of 1000 barrels of oil bought; x_2 be the number of barrels of aviation fuel processed in the cracker; x_3 be the number of barrels of heating oil processed (all variables are in unit of 1000 barrels). Then Linear Programming model can be formulated as following to maximize the overall profit.

$$\begin{aligned}
\text{maximize } & 60(0.5x_1 - x_2) + 130x_2 + 40(0.5x_1 - x_3) + 90x_3 - 40x_1 \\
\text{subject to } & x_1 \leq 20, \\
& 0.5x_1 - x_2 \geq 0, 0.5x_1 - x_3 \geq 0 \\
& x_2 + 0.75x_3 \leq 8 \\
& x_i \geq 0, i = 1, 2, 3
\end{aligned}$$

Optimal solution would be $(x_1, x_2, x_3) = (20, 8, 0)$, the corresponding maximized profit is 760.

Problem 2

Let x_i be the number of units of process i, $i = 1, 2$. (1 units of process 1 means we buy 1 unit of labor and 2 units of chemicals and get 3 oz of perfume by process 1. Analogously for 2.) Let x_3 denotes the number of hours to hire the promoter. Then the Linear programming can be formulated as,

$$\begin{aligned}
\text{maximize } & 3 * 5x_1 + 5 * 5x_2 - (1 * 3 + 2 * 2)x_1 - (2 * 3 + 3 * 2)x_2 - 100x_3 \\
\text{subject to } & x_1 + 2x_2 \leq 20000 \\
& 2x_1 + 3x_2 \leq 35000 \\
& 3x_1 + 5x_2 - 200x_3 \leq 1000 \\
& x_i \geq 0, i = 1, 2, 3
\end{aligned}$$

Optimal solution would be $(x_1, x_2, x_3) = (10000, 5000, 270)$, the corresponding maximized profit is 118000.

Problem 3

Let $x_i, i = 1, 2, 3$ be the number of barrels refined using method i. Let $y_{1j}, j = 1, 2, 3$ be the number of barrels of grade 6/8/10 oil correspondingly sold as gas and $y_{2j}, j = 1, 2, 3$ be the number of barrels of grade 6/8/10 oil correspondingly sold as heating oil. Let z_1 denote the number of barrels cracked from grade 6 to 8 and z_2 denote the number of barrels cracked from grade 8 to 10. The LP can be formulated as follows,

$$
\begin{aligned}
\text{maximize } & 12\sum_{j=1}^{3} y_{1j} + 5\sum_{j=1}^{3} y_{2j} - 3.4x_1 - 3x_2 - 2.6x_3 - z_1 - 1.5z_2 \\
\text{subject to } & 6y_{11} + 8y_{12} + 10y_{13} - 9\sum_{j=1}^{3} y_{1j} \geq 0 \\
& 6y_{21} + 8y_{22} + 10y_{23} - 7\sum_{j=1}^{3} y_{2j} \geq 0 \\
& \sum_{j=1}^{3} y_{1j} \leq 2000, \sum_{j=1}^{3} y_{2j} \leq 600 \\
& 0.3x_1 + 0.4x_2 + 0.1x_3 - y_{11} - y_{21} - z_1 \geq 0 \\
& 0.5x_1 + 0.2x_2 + 0.3x_3 - y_{12} - y_{22} + z_1 - z_2 \geq 0 \\
& 0.8x_1 + 0.4x_2 + 0.2x_3 - y_{13} - y_{23} + z_2 \geq 0 \\
& x_i \geq 0, i = 1, 2, 3, y_{ij} \geq 0, z_i \geq 0, i = 1, 2, j = 1, 2, 3
\end{aligned}
$$

$x^* = (1625, 0, 0), y_{1j} = (300, 400, 1300), y_{2j} = (187.5, 412.5, 0), z = (0, 0)$. The total revenue is 21475.

Problem 4

Let $x_i, i = 1, 2, ..., 10$ be the shares we sold for stock i. The problem can be formulated as below:

$$
\begin{aligned}
\text{maximize } & 36(100 - x_1) + 39(100 - x_2) + ... + 70(100 - x_{10}) \\
\text{subject to } & 0.99(30x_1 + 34x_2 + .. + 66x_{10}) - 0.3(10x_1 + 9x_2 + ... + x_{10}) \geq 30000 \\
& x_i \leq 100, i = 1, 2, ..., 10 \\
& x_i \geq 0, i = 1, 2, ..., 10
\end{aligned}
$$

The optimum policy is $x_3 = x_4 = x_8 = x_9 = x_{10} = 100, x_6 = 63.75$ and others are zero.

Problem 5

Decision variables.

- For $t = 1, \ldots, T-1$ and $i = 1, \ldots, n$, let $x_{i,t}$ be the amount borrowed from credit line i in month t.
- For $t = 1, \ldots, T$, let z_t be the amount available in cash in month t.

Objective function. $\max z_T$.

Constraints.

1. We require all variables to be nonnegative:

$$x, z \geq 0.$$

2. If a credit line require payments 3 months after the month when the money is borrowed, we cannot use it to borrow money in time $T-3$ or later:

$$x_{i,t} = 0 \text{ for all } i \text{ such that } r_{i,3} > 0 \text{ and } t = T-3, T-2, T-1.$$

Similarly, if a credit line require payments 2 (resp., 1) months after the when the money is borrowed, we cannot use it to borrow money in time $T-2$ or later (resp., $T-1$) :

$$x_{i,t} = 0 \text{ for all } i \text{ such that } r_{i,2} > 0 \text{ and } t = T-2, T-1.$$

$$x_{i,T-1} = 0 \text{ for all } i \text{ such that } r_{i,1} > 0.$$

3. Letting $\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}$ and for $t = 1, \ldots, T$, $x_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \\ \vdots \\ x_{n,t} \end{pmatrix}$, we bound the amount of money that can be borrowed at any time by each credit line by writing

$$x_t \leq \beta \quad \text{for } t = 1, \ldots, T.$$

4. We next write the constraints to satisfy the cash flow requirements at all times. For time $t = 1$, it can be written as

$$\sum_{i=1}^{n} x_{i,1} \ - \ z_1 = -c_1.$$

In time $t = 2$, we will have to pay $r_{i,1}$ for each dollar invested in credit line 1 made in time 1, and we may have some return from money invested in the money market account in the previous month. Thus, the corresponding cash-flow constraint is:

$$\sum_{i=1}^{n} x_{i,2} - \sum_{i=1}^{n} r_{i,1} x_{i,1} + (1+r) z_1 \ - \ z_2 = -c_2.$$

Similarly, at time $t = 3$ we have

$$\sum_{i=1}^{n} x_{i,3} - \sum_{i=1}^{n} r_{i,2} x_{i,1} - \sum_{i=1}^{n} r_{i,1} x_{i,2} + (1+r) z_2 \ - \ z_3 = -c_3.$$

At a generic time $t = 4, \ldots, T-1$ we have

$$\sum_{i=1}^{n} x_{i,t} - \sum_{i=1}^{n} r_{i,t-1} x_{i,t-1} - \sum_{i=1}^{n} r_{i,t-2} x_{i,t-2} - \sum_{i=1}^{n} r_{i,t-3} x_{i,t-3} + (1+r) z_{t-1} \ - \ z_t = -c_t.$$

Last, at time T, we cannot borrow any money, thus we have:

$$-\sum_{i=1}^{n} r_{i,T-1} x_{i,T-1} - \sum_{i=1}^{n} r_{i,T-2} x_{i,T-2} - \sum_{i=1}^{n} r_{i,T-3} x_{i,T-3} + (1+r) z_{T-1} \ - \ z_T = -c_T.$$